

PFAS results

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PFAS

Category and filtration

```
Family: Gamma ( log )
Formula:      concentration ~ category + adv_filtration
Zero inflation:      ~.
Data: pfas.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
514.4	528.1	-250.2	500.4	45

Dispersion estimate for Gamma family (σ^2): 0.35

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.5472	0.1233	44.99	< 2e-16 ***
categoryMunicipal	0.8109	0.2427	3.34	0.000835 ***
adv_filtrationYes	0.5391	0.4003	1.35	0.178101

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.8136	0.3801	-2.140	0.0323 *
categoryMunicipal	-3.1237	1.4194	-2.201	0.0278 *
adv_filtrationYes	2.9727	1.1173	2.660	0.0078 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Confidence intervals

Gamma components

	2.5 %	97.5 %	Estimate
cond.(Intercept)	201.45	326.64	256.52
cond.categoryMunicipal	1.40	3.62	2.25
cond.adv_filtrationYes	0.78	3.76	1.71

Zero-inflated components

	2.5 %	97.5 %	Estimate
zi.(Intercept)	4.75	1.07	2.26
zi.categoryMunicipal	367.12	1.41	22.73
zi.adv_filtrationYes	0.46	0.01	0.05

Detection in spring sources

A tibble: 2 x 5

source_type	num.detect	num.zero	total	percent
<fct>	<int>	<int>	<int>	<dbl>
1 Non-spring	0	13	13	0
2 Spring	23	4	27	85.2

Exact Two-Sample Fisher-Pitman Permutation Test

data: nd by factor(source_type) (Non-spring, Spring)

Z = 5.0404, p-value = 1.978e-07

alternative hypothesis: true mu is not equal to 0

Conclusions

```
pvals.pfas <- NULL
# Conclusion 1
# category, zero inflated component
pvals.pfas[1] <- 0.0278
# category, conditional component
pvals.pfas[2] <- 0.000835
# Conclusion 2
# source, fisher pitman
pvals.pfas[3] <- 1.978e-07
# Conclusion 3
# advanced filtration, zero inflated component
pvals.pfas[4] <- 0.0078
# advanced filtration, conditional component
pvals.pfas[5] <- 0.178101
names(pvals.pfas) <- c(
  "Conclusion.pfas1.1", "Conclusion.pfas1.2",
  "Conclusion.pfas2",
  "Conclusion.pfas3.1", "Conclusion.pfas3.2"
)
```

Conclusion.pfas1.1	Conclusion.pfas1.2	Conclusion.pfas2	Conclusion.pfas3.1
0.03	0.00	0.00	0.01
Conclusion.pfas3.2			
0.18			

PFAS Writeup

Conclusion 1: Bottled v. municipal

Conclusion 1:

- The odds of any PFAS being detected in a municipal water sample is 22.73 times higher compared to bottled water (adjusted ratio: 22.73, 95% interval: [1.41, 367.12]), keeping advanced filtration fixed. This confidence interval is quite wide.
 - Unadjusted p-value: 0.0278
 - Adjusted p-value: 0.03

- With 57% of bottled samples having PFAS detected, the probability of detection for municipal water is 1.68 higher than in bottled water (adjusted ratio: 1.68, 95% interval for relative risk: [1.14, 1.74]).
 - Unadjusted p-value: 0.0278
 - Adjusted p-value: 0.03
- For samples with PFAS detected, the mean PFAS concentration in a municipal water sample is 2.25 times higher compared to bottled water (adjusted ratio: 2.25, 95% interval: [1.4, 3.62])).
 - Unadjusted p-value: 0.000835
 - Adjusted p-value: 0
- Overall, the evidence suggests municipal water has more PFAS compared to bottled water.

Conclusion 2: Spring v. non-spring

Conclusion 2:

- We want to compare spring and non-spring sources. However, this is a challenging task. There are no municipal sources from spring water. Thus, if I tell you a sample is spring water, you know it is definitely bottled water. This makes it hard to disentangle the effect of the source from whether a sample is from bottled or municipal water. Thus, I want to restrict to looking at spring v. non-spring source for bottled water only. However, this still remains challenging—among non-spring bottled water, all of them have no PFAS detected! So, I should not be looking at mean concentration comparing spring and non-spring sources, because so many samples have 0 concentration. Thus, we only compare whether the proportion of samples with PFAS detected is different among spring and non-spring sources among bottled water.
- Among bottled water samples, the proportion of samples with PFAS detected is significantly higher in spring sources (85%, 23/27) compared to non-spring sources (0%, 0/13).
 - Unadjusted p-value: 1.978e-07
 - Adjusted p-value: 0
- There is strong evidence that in bottled water, spring sources are more likely to have PFAS detected.

Conclusion 3, advanced filtration

Conclusion 3:

- Among all samples, the odds of any PFAS being detected in a sample with advanced filtration is 95% lower compared to a sample without, keeping category fixed (adjusted ratio: 0.05, 95% interval for odds ratio: [0.01, 0.46]).
 - Unadjusted p-value: 0.0078
 - Adjusted p-value: 0.01
- With 76% of samples without advanced filtration having microplastics detected, the probability of detection for samples with advanced filtration is 82% lower than in samples without (adjusted ratio: 0.18, p-value: 95% interval for relative risk: [0.02, 0.78]).
 - Unadjusted p-value: 0.0078
 - Adjusted p-value: 0.01
- For samples with PFAS detected, the mean PFAS concentration in a sample with advanced filtration is 1.71 times higher compared to a sample without, but the difference is not significant (adjusted ratio: 1.71, 95% interval for multiplicative effect: [0.78, 3.76]).
 - Unadjusted p-value: 0.178101
 - Adjusted p-value: 0.18
- Overall, the evidence suggests there is little or no difference between having advanced filtration or not.